Intelligent Measurement Systems - Student Projects

General Context

All projects are based on a shared weather data REST API provided by the lecturer. The API exposes simulated weather measurements that can be used for data engineering, data analytics, machine learning, and visualization tasks.

The goal is to build end-to-end analytical solutions using weather data streams. Each team should design a small data pipeline, collect data from the API, process it, analyze it, and present the results.

Shared Weather REST API

Base URL

```json
https://e6uw49pbah.execute-api.us-east-1.amazonaws.com/dev
```

Authentication

Authorization: Bearer STUDENT_TOKEN_2026

Requests without the token will return

```json
{
  "message": "Unauthorized. Missing or invalid token."
}
```

Available API Endpoints

Get Latest Weather Measurement

```json
GET /weather/latest?station_id=GDN_01
```

Example

```
JSON
curl -X GET \
  "https://e6uw49pbah.execute-api.us-east-1.amazonaws.com/dev/weather/latest?station_id=GDN_01" \
  -H "Authorization: Bearer STUDENT_TOKEN_2026"
```

Example response:


```
JSON
{
  "timestamp": "2026-05-05T20:29:01.406116+00:00",
  "station_id": "GDN_01",
  "temperature": 19.99,
  "humidity": 79.62,
  "pressure": 1008.92,
  "wind_speed": 10.87,
  "wind_direction": 173,
  "rain_mm": 0,
  "cloud_cover": 67
}
```


Get Batch of Weather Measurements

```
JSON
GET /weather/batch?station_id=GDN_01&limit=100
```

Example:

```
JSON
curl -X GET \
  "https://e6uw49pbah.execute-api.us-east-1.amazonaws.com/dev/weather/batch?station_id=GDN_01&limit=100" \
  -H "Authorization: Bearer STUDENT_TOKEN_2026"
```

This endpoint should be used when the project requires more data for analysis, model training, visualization, or historical comparison.

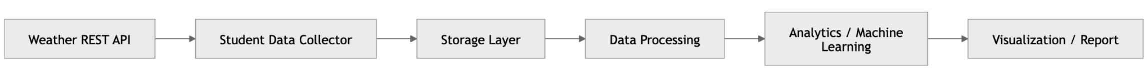
Get Available Weather Stations

```json
GET /weather/stations
```

Example:

```json
curl -X GET \
"https://e6uw49pbah.execute-api.us-east-1.amazonaws.com/dev/weather/stations" \
  -H "Authorization: Bearer STUDENT_TOKEN_2026"
```

General Architecture



Possible storage options:

- local CSV / JSON files,
- AWS S3,
- DynamoDB,
- relational database,
- data lake structure on S3.

Possible analytical tools:

- Python,
- pandas,
- scikit-learn,
- PySpark,
- AWS Glue,
- Athena,
- QuickSight,
- SageMaker,
- Jupyter Notebook.

Project 1 - Temperature Anomaly Detection

Goal

The goal of this project is to detect unusual temperature behavior in a stream of weather measurements.

Students should build a data pipeline that collects weather data from the REST API, stores it, and analyzes whether temperature values are normal or anomalous.

An anomaly can be defined as:

- a sudden temperature jump,
- a value outside a normal range,
- a value significantly different from recent observations,
- a value that does not fit the expected daily pattern.

Suggested Analytical Methods

Students may use:

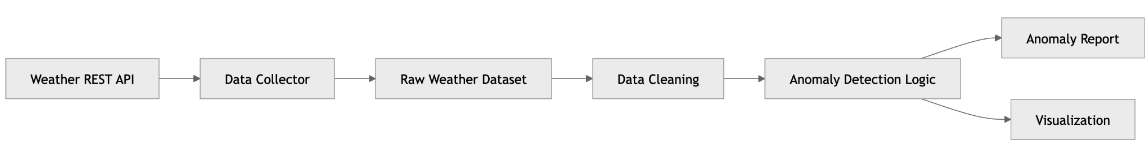
- moving average,
- rolling standard deviation,
- Z-score,
- interquartile range,
- rule-based anomaly detection,
- simple machine learning model.

Expected Output

The project should produce:

- a list of detected anomalies,
- charts showing normal and anomalous points,
- explanation of the detection method,
- final conclusions.

Diagram



Example Questions

- When did the most unusual temperature changes occur?
- How sensitive is the anomaly detection method?
- How many false positives can be observed?
- Does the anomaly detection method work better on raw or aggregated data?

Project 2 - Temperature Forecasting

Goal

The goal of this project is to build a simple forecasting model that predicts future temperature values based on historical weather measurements.

Students should collect a dataset from the API, prepare features, train a model, and evaluate prediction quality.

Suggested Features

Possible input features:

- previous temperature values,
- hour of day,
- humidity,
- pressure,
- wind speed,
- cloud cover,
- rain level.

Suggested Models

Students may use:

- linear regression,
- random forest regression,
- gradient boosting,
- moving average baseline,
- simple neural network,
- time-series model.

Expected Output

The project should produce:

- trained forecasting model,
- prediction results,
- comparison between predicted and actual values,
- error metrics such as MAE, RMSE, or MAPE,
- visualization of forecast quality.

Diagram

Example Questions

- Which features improve prediction quality?
- Is pressure useful for forecasting temperature?
- How does a machine learning model compare with a moving average baseline?
- How many historical observations are needed for acceptable results?

Project 3 - Weather Comfort Index

Goal

The goal of this project is to calculate and visualize a weather comfort index based on weather conditions.

The system should classify weather conditions as comfortable or uncomfortable for selected activities, such as walking, running, cycling, or outdoor learning.

Input Variables

Students may use:

- temperature,
- humidity,
- wind speed,
- rain level,
- cloud cover.

Suggested Logic

The comfort score can be rule-based, for example:

```None
High comfort:

temperature between 16°C and 24°C,

low rain,

moderate wind,

humidity below 75%.
```

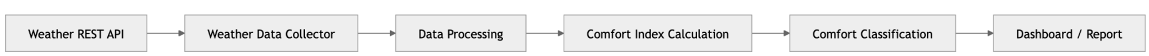
Or it can be calculated as a numerical score from 0 to 100.

Expected Output

The project should produce:

- comfort score,
- comfort class,
- dashboard or report,
- recommendation for outdoor activity.

Diagram



Example Questions

- What are the best weather conditions for outdoor activity?
- Which parameter has the strongest negative impact on comfort?
- How does comfort change during the day?
- Can different activities have different comfort thresholds?

Project 4 - Weather Alerting System

Goal

The goal of this project is to build an alerting system that detects potentially dangerous weather conditions.

The system should collect weather data from the API and trigger alerts when predefined thresholds are exceeded.

Example Alert Rules

Possible alert conditions:

- wind speed above selected threshold,
- sudden pressure drop,
- heavy rainfall,
- very low temperature,
- very high temperature,
- high humidity combined with high temperature.

Suggested AWS Services

Students may use:

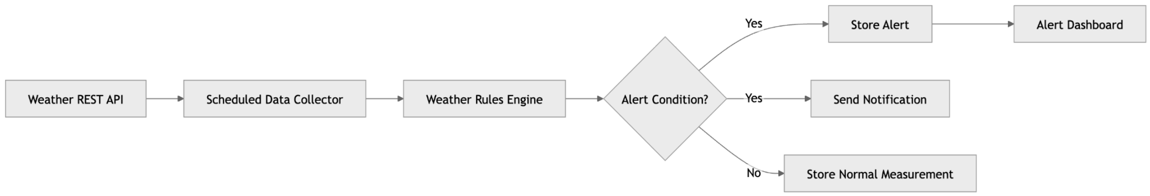
- Lambda,
- EventBridge,
- SNS,
- DynamoDB,
- CloudWatch,
- S3.

Expected Output

The project should produce:

- rule engine for weather alerts,
- alert log,
- optional email/SNS notification,
- dashboard of alert history.

Diagram



Example Questions

- Which weather conditions should generate alerts?
- How often should data be checked?
- How can false alerts be reduced?
- Should alerts be based on single values or trends?

Project 5 — Weather Type Classification

Goal

The goal of this project is to classify weather conditions into categories based on measurement data.

Example classes:

- sunny,
- cloudy,
- rainy,
- windy,
- cold,
- warm,
- unstable,
- storm-risk.

Suggested Approaches

Students may implement:

- rule-based classification,
- machine learning classification,
- comparison between rule-based and ML-based classification.

Example Rules

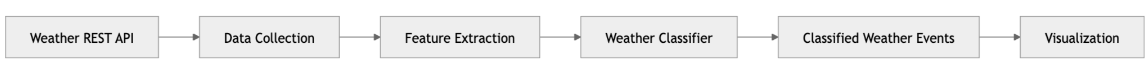
```
None
Rainy:

rain_mm > 0.5 and cloud_cover > 70

Windy:

wind_speed > 10

Cloudy:

cloud_cover > 80 and rain_mm == 0

Warm:

temperature > 22
```

Expected Output

The project should produce:

- weather classification logic,
- classified dataset,
- visual summary of weather types,
- confusion matrix if ML is used,
- explanation of classification strategy.

Diagram



Example Questions

- Which class occurs most often?
- Are rule-based classifications reliable?
- Which variables are most important for classification?
- Can machine learning outperform simple rules?

Project 6 - Weather Data Quality Monitoring

Goal

The goal of this project is to monitor and evaluate the quality of weather data.

Students should build a system that checks whether incoming weather measurements are complete, valid, consistent, and reasonable.

Example Data Quality Checks

Students may check:

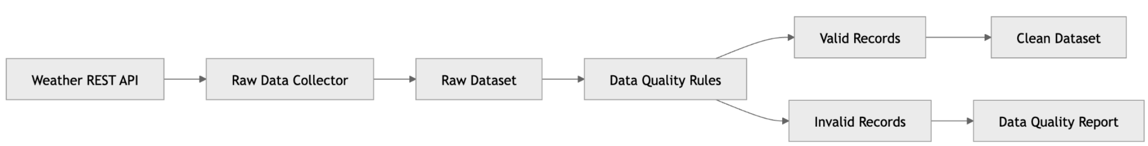
- missing values,
- duplicated records,
- invalid timestamp,
- timestamp from the future,
- humidity outside 0–100 range,
- pressure outside expected range,
- negative wind speed,
- unrealistic temperature jump,
- inconsistent rain and cloud cover values.

Expected Output

The project should produce:

- validation rules,
- data quality report,
- quality score,
- list of invalid records,
- recommendation on how to clean the data.

Diagram



Example Questions

- What percentage of records is valid?
- Which quality issue is most common?
- How should invalid records be handled?
- Can data quality be monitored automatically?

Project 7 - Multi-Station Weather Comparison

Goal

The goal of this project is to compare weather conditions between multiple weather stations.

Students should collect data for different station IDs and analyze differences between locations.

Example stations:

```None
GDN_01

GDN_02

GDY_01

SOP_01
```

Suggested Analysis

Students may compare:

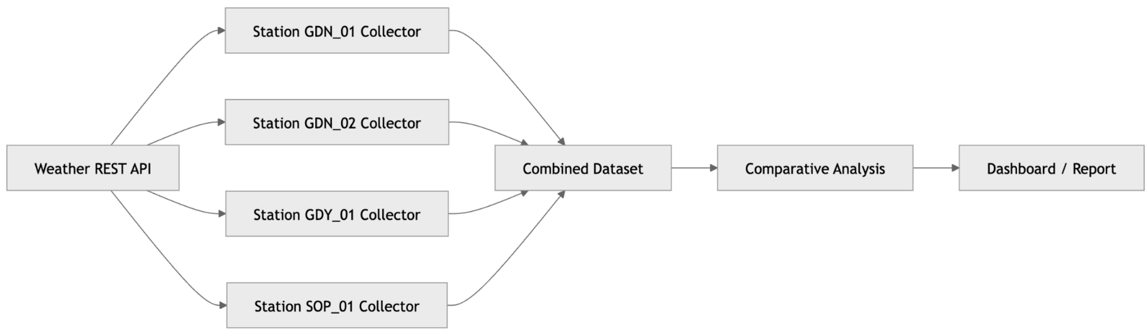
- average temperature,
- wind speed,
- rainfall,
- pressure,
- humidity,
- comfort index,
- anomaly frequency.

Expected Output

The project should produce:

- multi-station dataset,
- comparative analysis,
- charts showing station differences,
- explanation of possible microclimate patterns.

Diagram



Example Questions

- Which station has the highest average temperature?
- Which station is the windiest?
- Are station measurements correlated?
- Can we detect location-specific weather patterns?

Project 8 - Mini Lakehouse for Weather Data

Goal

The goal of this project is to build a small data engineering pipeline inspired by lakehouse architecture.

Students should organize weather data into multiple data layers:

- Bronze — raw data,
- Silver — cleaned and validated data,
- Gold — aggregated analytical data.

Suggested Technologies

Students may use:

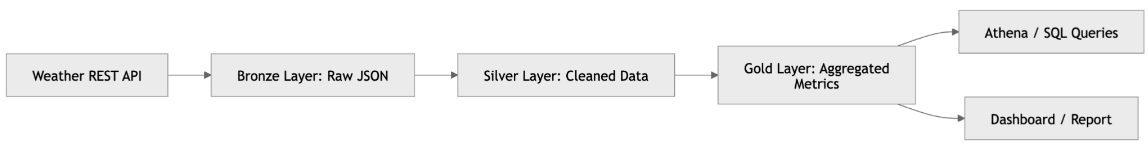
- AWS S3,
- AWS Glue,
- Athena,
- Python,
- PySpark,
- pandas,
- DuckDB,
- dbt optional.

Expected Output

The project should produce:

- raw weather data storage,
- cleaned dataset,
- aggregated metrics,
- SQL queries or analytical notebooks,
- documentation of data layers.

Diagram



Example Gold Metrics

Students may calculate:

- average temperature per station,
- maximum wind speed per day,
- total rainfall per station,
- average humidity per hour,
- number of alerts per station,
- comfort index per day.

Example Questions

- Why do we separate raw, cleaned, and aggregated data?
- What data quality rules should be applied in the Silver layer?
- Which aggregations are useful for reporting?
- How can the pipeline be automated?